NFL Graphs - C

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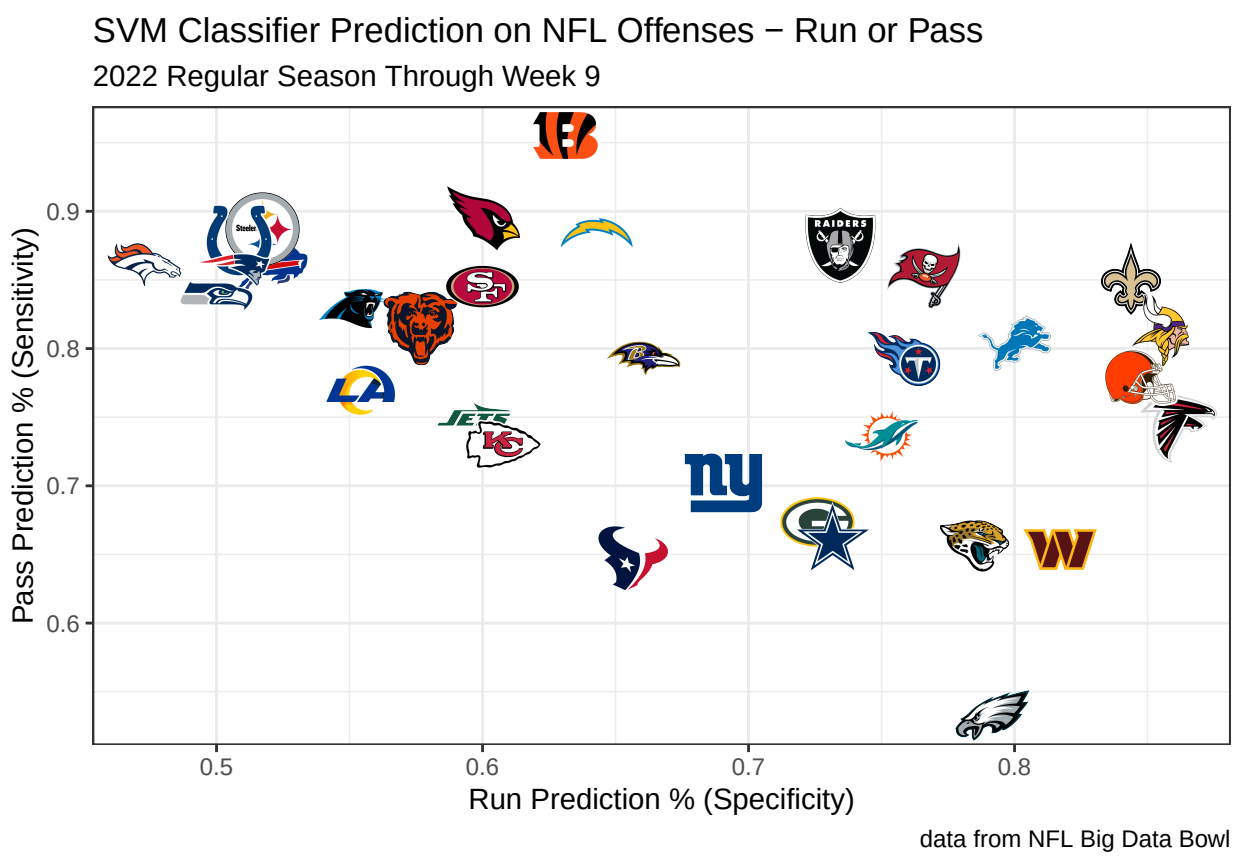
2024-12-03

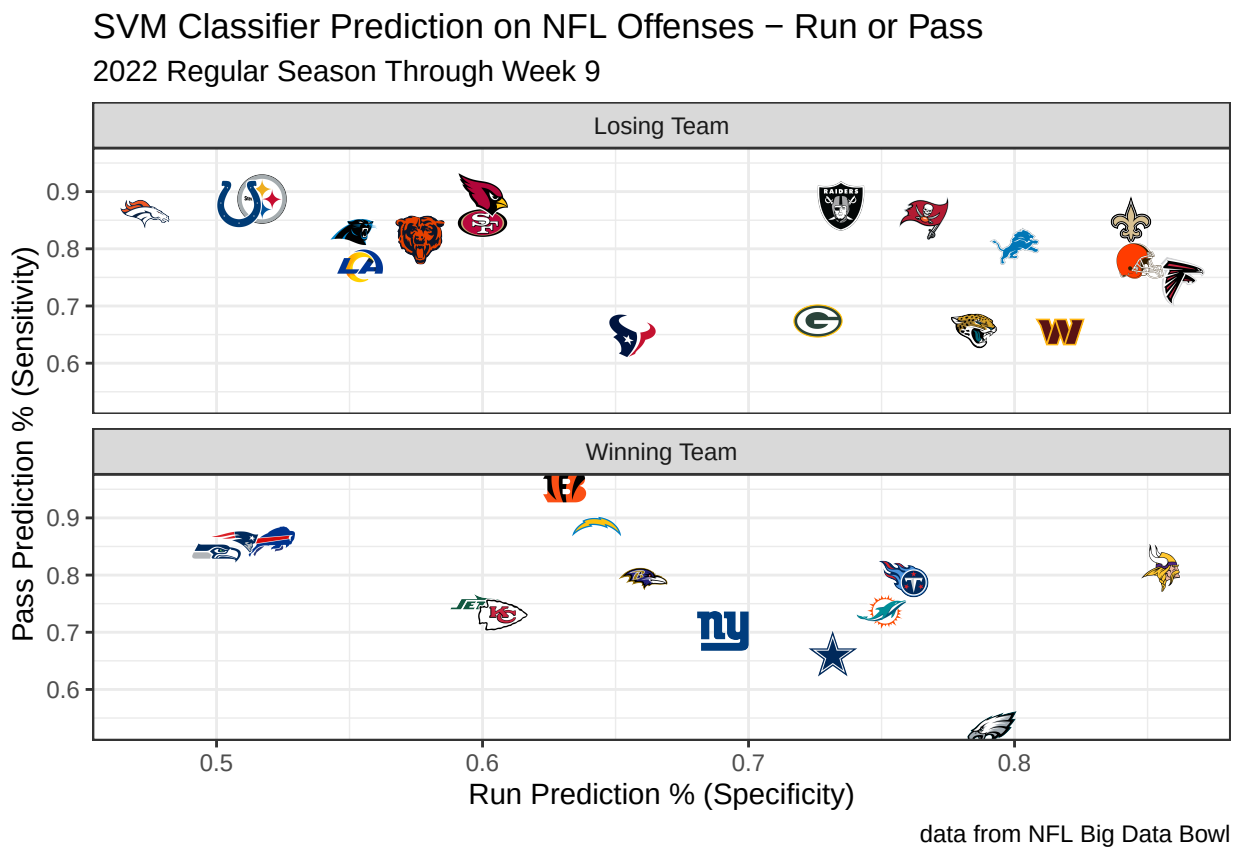
Positive class = "pass" Negative class = "run" Sensitivity refers to pass Specificity refers to run

```
# this number will scale the image sizes proportionally for testing
s <- 1.25

svm_data %>%
  ggplot(aes(y = sens, x = spec)) +
  geom_nfl_logos(aes(team_abbr = team, width = 0.075 * s, height = 0.1 * s)) +
  labs(title = "SVM Classifier Prediction on NFL Offenses - Run or Pass",
       subtitle = "2022 Regular Season Through Week 9",
       y = "Pass Prediction % (Sensitivity)",
       x = "Run Prediction % (Specificity)",
       caption = "data from NFL Big Data Bowl") +
  theme_bw()
```

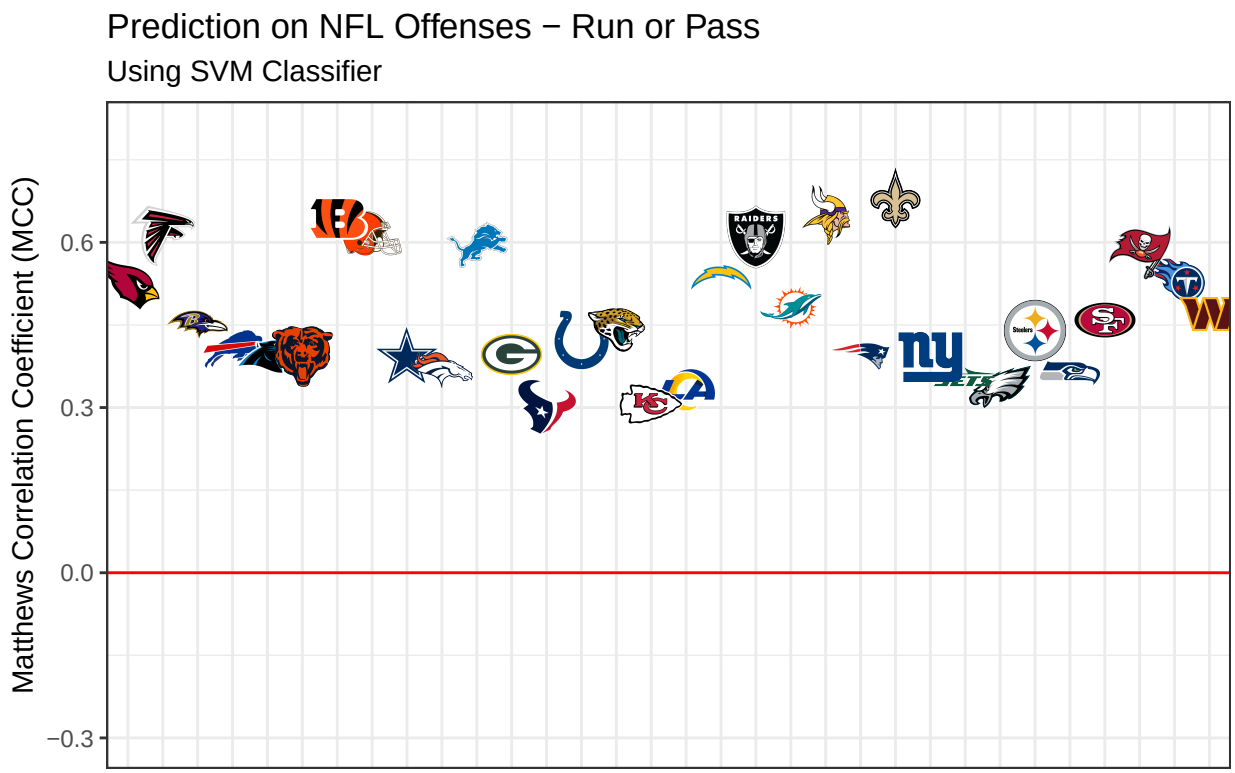


```r
s <- 2
svm_data %>%
  ggplot(aes(y = sens, x = spec)) +
  facet_wrap(wining_record~., nrow = 2) +
  geom_nfl_logos(aes(team_abbr = team, width = 0.075 * s, height = 0.1 * s)) +
  labs(title = "SVM Classifier Prediction on NFL Offenses - Run or Pass",
       subtitle = "2022 Regular Season Through Week 9",
       y = "Pass Prediction % (Sensitivity)",
       x = "Run Prediction % (Specificity)",
       caption = "data from NFL Big Data Bowl") +
  theme_bw()
```




```r
s <- 1
svm_data %>%
  ggplot(aes(y = mcc, x = team)) +
  geom_nfl_logos(aes(team_abbr = team, width = 0.075 * s, height = 0.1 * s)) +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Prediction on NFL Offenses - Run or Pass",
       subtitle = "Using SVM Classifier",
       y = "Matthews Correlation Coefficient (MCC)",
       x = "",
       caption = "data from NFL Big Data Bowl: 2022 regular season weeks1-9") +
  scale_y_continuous(limits = c(-0.3, 0.8)) +
  theme_bw() +
  theme(axis.ticks.x = element_blank(),
```

```
        axis.text.x = element_blank())
```




```
svm_data %>% select(team, wins, mcc) %>%
  arrange(mcc) %>% slice(1:5) %>% kable()
```

team	wins	mcc
HOU	1	0.3004505
KC	6	0.3065196
LA	3	0.3319766
PHI	8	0.3371568
NYJ	6	0.3509050